

- E waste of Digital IC :- Old or discarded electronic gadgets such as phone, computers, television may contain memory chip, microprocessor and other semiconductor devices they called E waste of Digital IC.
- Improper Disposal of Digital IC can have environmental and health issues.
- For Responsible ewaste management of Digital ICs it is essential to adopt proper disposal method.  
This includes.
  - (1) Recycling :- Digital IC can be recycled to recover valuable materials such as gold, copper and silver. Recycling facilities can extract these materials and Resuse them in the production of new electronic components.
  - (2) Proper storage :- ICs should be storage in a Secure and Dry environment to prevent damage and contamination.
  - (3) Proper disposal :- Digital ICs should be disposed off through authorized e-waste recycling facilities or Designated collection points.

## Best Practices:-

- Design for Sustainability :- Designing electronic products with sustainability in mind can reduce electronic waste and promote responsible disposal practices.
- Extended Producer Responsibility :- Manufacturers should take responsibility for the disposal of their products and ensure that they are properly recycled or disposed of.
- Public awareness :- Educating the public about the importance of e-waste management, health and environment issues can promote responsible behaviour and encourage individuals to take action.

Q:- Compare SRAM & DRAM

| Static RAM                                                 | Dynamic RAM                                                                         |
|------------------------------------------------------------|-------------------------------------------------------------------------------------|
| > SRAM uses transistor to store a single bit of data       | > DRAM uses a separate capacitor to store each bit of data                          |
| > SRAM does not need periodic refreshment to maintain data | > DRAM needs periodic refreshment to maintain the charge in the capacitors for data |
| > SRAM's structure is complex than DRAM                    | > DRAM's structure is simplex than SRAM                                             |
| > SRAM are expensive as compared to DRAM                   | > DRAM's are less expensive as compared to SRAM                                     |
| > SRAM are faster than DRAM                                | > DRAM's are slower than SRAM                                                       |
| > SRAM are used in Cache memory                            | > DRAM are used in Main memory                                                      |

Q:- Compare RAM & ROM

| Parameter      | RAM                                               | ROM                                                   |
|----------------|---------------------------------------------------|-------------------------------------------------------|
| (1) Full Form  | Random Access memory                              | Read only memory                                      |
| (2) Read/Write | Read & Write both operation supported.            | only Read operation supported.                        |
| (3) Volatility | Volatile memory - when power off, data not Retain | Non-volatile memory<br>Data Retain even if power off. |
| (4) Speed      | High                                              | Slow                                                  |
| (5) Cost       | more                                              | Less                                                  |

## Comparison of different Logic family.

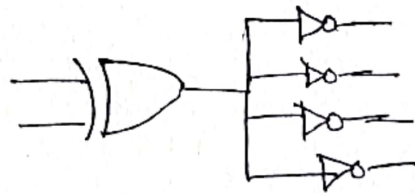
| Logic Family<br>→<br>Parameter          | DTL  | TTL     | ECL    | I <sup>2</sup> L | MOS | CMOS     |
|-----------------------------------------|------|---------|--------|------------------|-----|----------|
| (1) Basic Gate                          | Nand | Nand    | Nor/or | Nor              |     | Nor/Nand |
| (2) Noise Immunity                      | Good | V. Good | Poor   | Poor             |     | V. Good  |
| (3) Power dissipation<br>Per Gate in mW | 5-12 | 10      | 50     | 0.1              | 0.1 | 0.01     |
| (4) Propagation Delay<br>Time in ns     | 30   | 9       | 1      | 1                | 50  | < 50     |
| (5) Fan out                             | 5    | 10      | 10     | 8                | 10  | 50       |
| (6) Cost.                               | Low  | Low     | High   | Very Low         | Low | Low      |

① પ્રીયાગરેશન ડીલે :- લોજિક ગેટ ને ઇનપુટ પલ્સ આપવા માં આવે અને આઉટ પુર મળે તે બન્ને વચ્ચેના સમયગાળા ને પ્રીયાગરેશન ડીલે કહે છે. પ્રીયાગરેશન ડીલે ઓછો હોય સ્પીડ વધારે.

② ફેન ઇન :- ડિજિટલ લોજિક ગેટ ને વધુ મા વધુ જોડવા ઇનપુટ આપી શકાય નહોતો હોય ફેન ઇન કહે છે. નીચેના ગેટ ને 3 ઇનપુટ આપી શકાય છે તેથી તેનું Fan in 3 છે.



③ ફેન આઉટ :- આપેલા ગેટ આઉટ પુર વધુ મા વધુ જોડવા ગેટના ઇનપુટને ડ્રાઇવ કરી શકે છે તેને પ્રથમ ગેટનું ફેન આઉટ કહે છે.



EOR ગેટનું ફેન આઉટ = 4

Fan out = 4

④ પાવર ડિસીપેશન :- ડિજિટલ સર્કીટ જે કિમીના પાવર વાપર વાપરે છે તેને પાવર ડિસીપેશન કહે છે.

⑤ નોઈસ માર્જીન :- લોજિક ડિવાઈસ પ્રોત્તાનું એકચ્યુલ આઉટપુટ માં ફેરફાર થવા વગર વધુ મા વધુ જોડવા નોઈસ સહન કરી શકે તેને નોઈસ ઇમ્યુનીટી કહે છે. નોઈસ ઇમ્યુનીટી ના આકાશીય કીમા ને નોઈસ માર્જીન કહે છે.